

LIANG (DIVIN) YAN

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EDUCATION

- University of Washington, Incoming PhD in Computer Science** 2026.09–2030.06
- Paul G. Allen School of Computer Science & Engineering
- Fudan University, M.S. in Applied Mathematics** 2021.09–2025.06
- Institute of Science and Technology for Brain-Inspired Intelligence (ISTBI)
 - **GPA:** 3.51/4.00 (Top 3%)
 - **Selected Graduate Courses:** Neural Networks and Deep Learning (A), Causal Inference (A), Introduction to Algorithms (A-), Advanced Mathematical Statistics (B+), Reinforcement Learning (B+)
 - **Advisor:** [Prof. Zengfeng Huang](#)
- University of California, Merced, Visiting Graduate Student in Computer Science** 2023.07–2024.02
- Department of Electrical Engineering and Computer Science
 - **Advisors:** [Prof. Ming-Hsuan Yang](#), [Prof. Qi Lu](#)
- Jiangxi University of Finance and Economics, B.Econ. in (Economic) Statistics** 2017.09–2021.06
- School of Statistics and Data Science (**Honors Program: Experimental Class for Top Talents in Statistics**)
 - **GPA:** 92.17/100
 - **Selected Undergraduate Courses:** Mathematical Analysis (90/100), Advanced Algebra (97/100), Bayesian Decision (98/100), Real Analysis and Functional Analysis (97/100), Data Mining (93/100)
 - **Advisor:** [Prof. Xiaohui Liu](#)

RESEARCH EXPERIENCE

- California Institute of Technology** 2023.11–2025.06
- Visiting Student; research on generative models on Riemannian manifolds and physics-informed generative models for materials design, drug discovery, and physics.
 - **Advisor:** [Prof. Anima Anandkumar](#)
- University of California, Merced** 2023.07–2024.02
- Research Assistant and Visiting Student; generative models.
 - **Advisors:** [Prof. Ming-Hsuan Yang](#), [Prof. Lu Qi](#)
- Tencent AI Lab** 2022.04–2022.12
- Machine Learning Research Intern; generative AI for structure-based drug design.
 - **Advisor:** [Dr. Tingyang Xu](#)
- Shanghai Artificial Intelligence Laboratory** 2022.01–2022.03
- Machine Learning Research Intern; generative AI for science.
 - **Advisor:** [Dr. Xun Qian](#)
- Fudan University** 2021.09–2025.06
- Student Researcher; generative models, graph neural networks, and multimodal (diffusion) LLMs for science.
 - **Advisor:** [Prof. Zengfeng Huang](#)

PUBLICATIONS

(* Co-first authors; author order decided by a coin flip.)

- [1] **Manifold-Constrained Nucleus-Level Denoising Diffusion Model for Structure-Based Drug Design.**
*Shengchao Liu** (co-first), *Liang Yan** (co-first), *Weitao Du*, *Weiyang Liu*, *Hongyu Guo*, *Christian Borgs*, *Jennifer Chayes*, *Anima Anandkumar*.

- Proceedings of the National Academy of Sciences (PNAS), 2025. (Presented at ICML 2024 GRaM Workshop.)*
[\[Reported by Caltech News\]](#) [\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#)
- [2] **MGB: The Material Generation Benchmark.**
Liang Yan, Beom Seok Kang, Maurice D. Hanisch, Jian Ma, Anima Anandkumar.
NeurIPS 2025 AI4Mat Workshop.
[\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#) [\[Poster\]](#)
- [3] **A Comprehensive Study of Multimodal LLMs for Scientific Discovery with Benchmarking.**
Liang Yan, Xu Jiang, Jian Ma, Yuhang Liu, Tian Bian, Qichao Wang, Abhishek Basu, Yu Rong, Tingyang Xu, Pengcheng Wu, Le Song, Imran Razzak, Junchi Yan, Zengfeng Huang, Yutong Xie.
Under review at NeurIPS 2026. (Presented at NeurIPS 2025 VLM4RWD Workshop.)
[\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#)
- [4] **Geometric Imbalance in Semi-Supervised Node Classification.**
Liang Yan, Shengzhong Zhang, Bisheng Li, Menglin Yang, Chen Yang, Min Zhou, Weiyang Ding, Yutong Xie, Zengfeng Huang.
NeurIPS 2025. (Presented at ICML 2025 DataWorld Workshop.)
[\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#)
- [5] **Rethinking Semi-Supervised Imbalanced Node Classification from Bias-Variance Decomposition.**
Liang Yan, Gengchen Wei, Chen Yang, Shengzhong Zhang, Zengfeng Huang.
NeurIPS 2023.
[\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#)
- [6] **An Adaptive Autoregressive Diffusion Approach to Design Active Humanized Antibody and Nanobody.**
Jian Ma, Fandi Wu, Tingyang Xu, Shaoyong Xu, Wei Liu, **Liang Yan**, Minghao Qu, Xiaoke Yang, Qifeng Bai, Junyu Xiao, Jianhua Yao.
Nature Machine Intelligence, 2025.
[\[Paper\]](#) [\[Arxiv\]](#) [\[Code\]](#)
- [7] **A Multi-Grained Symmetric Differential Equation Model for Learning Protein-Ligand Binding Dynamics.**
Shengchao Liu, Weitao Du, Hannan Xu, Yanjing Li, Zhuoxinran Li, Vignesh Bhethanabotla, **Liang Yan**, Christian Borgs, Anima Anandkumar, Hongyu Guo, Jennifer Chayes.
Nature Communications, 2025. (Earlier version: ICLR 2024 AI4DifferentialEquations Workshop (Oral).)
[\[Project Page\]](#) [\[Paper\]](#) [\[Code\]](#)
- [8] **Hierarchical Graph Latent Diffusion Model for Conditional Molecule Generation.**
Tian Bian, Yifan Niu, Heng Chang, **Liang Yan**, Tingyang Xu, Yu Rong, Jia Li, Hong Cheng.
CIKM 2024.
[\[Paper\]](#) [\[OpenReview\]](#)
- [9] **A Geometric Foundation Model for Crystalline Material Discovery.**
Shengchao Liu* (co-first), **Liang Yan*** (co-first), Weitao Du, Zhuoxinran Li, Zhiling Zheng, Omar M. Yaghi, Christian Borgs, Hongyu Guo, Anima Anandkumar, Jennifer T. Chayes.
NeurIPS 2024 AI4Mat Workshop.
[\[Paper\]](#) [\[Code\]](#)
- [10] **Training Class-Imbalanced Diffusion Model via Overlap Optimization.**
Liang Yan, Lu Qi, Vincent Tao Hu, Ming-Hsuan Yang, Meng Tang.
ArXiv 2024.
[\[Paper\]](#) [\[Code\]](#)
- [11] **An Equivariant Flow Matching Framework for Learning Molecular Crystallization.**
Shengchao Liu, **Liang Yan**, Hongyu Guo, Anima Anandkumar.
ICML 2024 GRaM Workshop; ICML 2024 ML4LMS Workshop.
[\[Paper\]](#) [\[Code\]](#)
- [12] **Physics Informed Spectral Element Network with Positional Parameter.**
Gengchen Wei, **Liang Yan**, Lei Bai, Wanli Ouyang, Chen Lin.
ArXiv 2023.
[\[Paper\]](#)

RESEARCH INTERESTS

- | | |
|--|-----------------|
| Large Language Models | 2025.12–Present |
| <ul style="list-style-type: none"> • <i>Architectural Foundations</i>: Non-Transformer LLM architecture design. • <i>Agent</i>: memory reasoning and retrieval agent system. • <i>Science of LLMs</i>: explainability, phase transition. | |
| Generative Models | 2023.11–Present |
| <ul style="list-style-type: none"> • <i>Representation Learning</i>: geometric deep learning with physical priors, symmetry learning, SE(3)-equivariance and invariance, and group-theoretic methods. • <i>Generative Models</i>: diffusion and flow-based generative models, stochastic and ordinary differential equations | |

(SDEs/ODEs), annealed Langevin dynamics, optimal transport, and probability-flow ODEs.

- *Riemannian Manifold*: generative modeling and representation learning on Riemannian manifolds.

AI for Science and Biomedicine

2023.11–Present

- *Materials science*: generative AI and multimodal LLMs for MOF and crystal-structure prediction, de novo crystal generation, disordered-material design, space-group–constrained modeling, and quantum-materials design.
- *Drug discovery and chemistry*: structure-based drug design (SBDD), antibody and nanobody design, and spatial transcriptomics.
- *Physics*: AI for density functional theory (DFT), Hamiltonian prediction, and solving partial differential equations (PDEs).

Foundations and theory of non-Euclidean neural networks

2022.05–Present

- *Robustness, generalization, and fairness*: fundamental learning algorithms for robust and fair non-Euclidean models under data bias and distribution shifts.

AWARDS

- Academic Scholarship of Fudan University (Top 3%, 2022-2023)
- Academic Scholarship of Fudan University (Top 3%, 2021-2022)
- Bayi Sunshine Scholarship of Jiangxi Province (Top 1%, 2017-2018)

INVITED AND HOSTED TALKS

- Nov. 2022: Imbalance Learning on Graphs. *NOAH'S ARK LAB*, host: *Dr. Min Zhou*.
- Nov. 2024: Optimal Transport and Schrödinger Bridge for Molecular Design. *Tencent AI Lab*, host: *Dr. Yu Rong*.

SERVICE

- Reviewer: NeurIPS (2023-2026), ICLR (2024-2026), ICML (2024-2025), KDD (2023-2024), ACM MM (2025-2026), ACM MM Datasets Track (2025-2026).

SKILLS

- Python, Pytorch, PYG, R, Stata, Latex, Git, Linux.
- Basketball. Former Captain of the College Basketball Team; Member of the University Basketball Team and an Amateur Basketball Club.